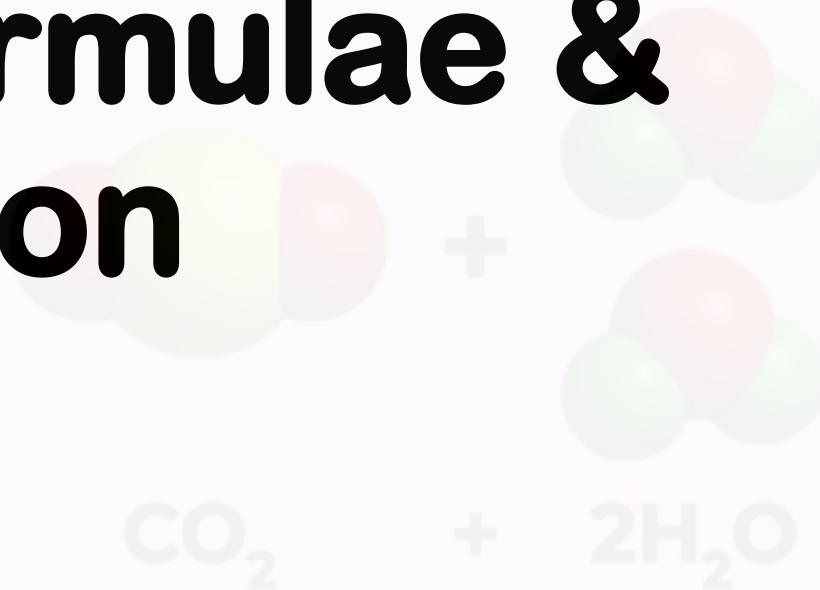


Chemical Formulae & Equation

Reactants
(The starting materials)



Products
(The newly formed substances)



Course Outline

- Chemical reactions and chemical equations
- General properties of aqueous solution
- The Mole & Avogadro's Numbers Molar volume of gases
- Empirical and molecular formulae
- Concentration of solutions
- Stoichiometry of reaction in solution
- Acid-base reactions
- Oxidation and reduction reactions
- Limiting reagents
- Reaction yields

Learning Objectives

Foundations & The Mole Concept

Balance chemical equations using coefficients
Define & apply the mole concept (Find average atomic mass & molar volume of gases)
Calculate formulas (empirical vs. molecular)

Solutions & Stoichiometry

Distinguish between solutes, solvents, and solutions
Calculate concentrations of regular and diluted solutions
Master solution stoichiometry in chemical reactions
Classify electrolytes (strong, weak, or non-electrolytes)

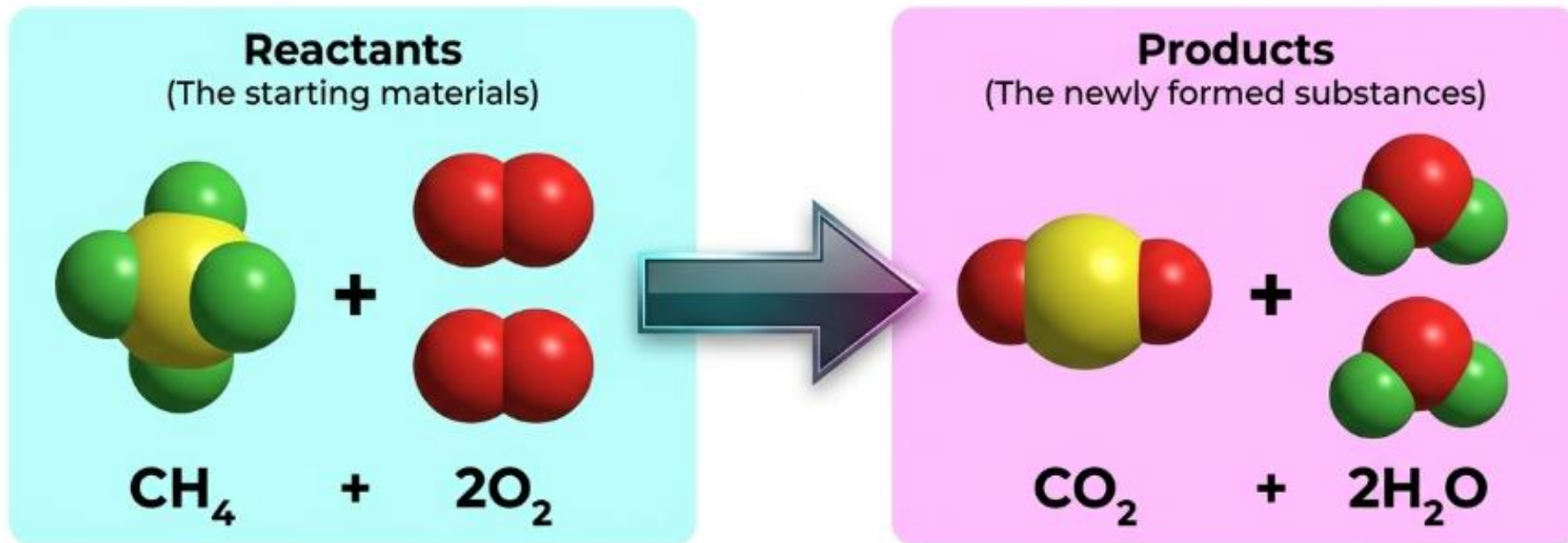
Reaction Yields & Limitations

Identify limiting reactants in a chemical reaction
Calculate reaction yields (theoretical, actual, and percentage yield)

Acids, Bases, & Redox Reactions

Compare acid-base theories (Arrhenius vs. Brønsted-Lowry)
Classify acids & bases (strong vs. weak) and identify conjugate pairs
Summarize neutralization reactions
Analyze redox reactions (oxidation numbers, agents, and half-reactions)

Chemical Change



Atoms are neither created nor destroyed. A chemical reaction is simply the reorganization of atoms into new substances.

Chemical Reaction & Chemical Equation

- ❑ A chemical reaction is a process in which one or more substances are changed into one or more new substances.
- ❑ A chemical equation uses chemical symbols to show what happens during a chemical reaction.

Information Provided by Chemical Equations

- Nature of the reactants and products
 - Compounds involved in the reaction
 - Physical states of the reactants and products

State	Symbol
Solid	(<i>s</i>)
Liquid	(<i>l</i>)
Gas	(<i>g</i>)
Dissolved in water (in aqueous solution)	(<i>aq</i>)

Information Provided by Chemical Equations

- Relative numbers of reactants and products
 - Indicated by coefficients in a balanced equation
- Mass remains constant
 - Atoms are conserved in a chemical reaction

Writing a Chemical Equation



The Micro View

2 atoms Mg + 1 molecule O₂ makes 2 formula units MgO

The Molar View

2 moles Mg + 1 mole O₂ makes 2 moles MgO

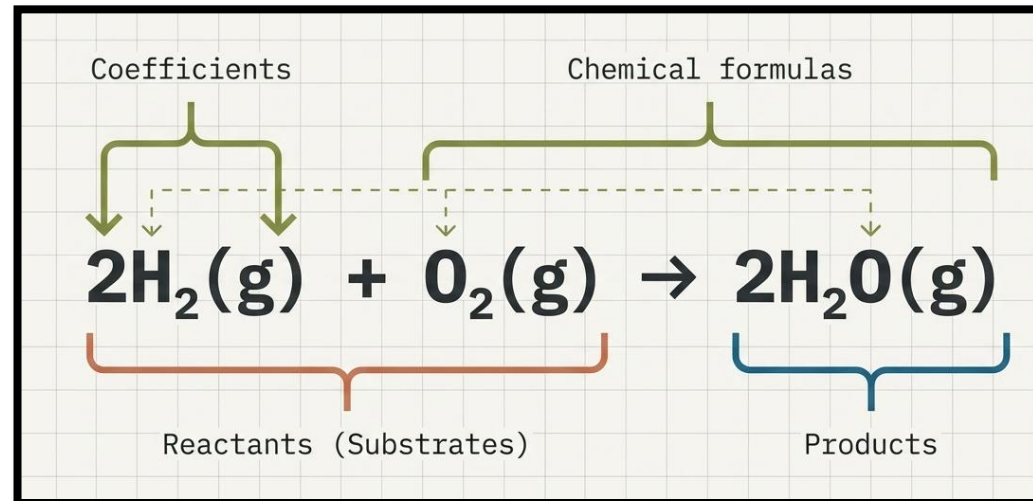
The Lab View

48.6 grams Mg + 32.0 grams O₂ makes 80.6 grams MgO

Notice that mass remains constant across the reaction. The equation tells us the **exact proportions** needed, just like a baking recipe.

Balancing a Chemical Equation

- All atoms present in the reactants must be accounted for among the products
 - Same number of each type of atom should be present on the products side and on the reactants side of the arrow



Things to Remember

- An unbalanced equation is of limited use
- Identities of the reactants and products of a reaction must be recognized by experimental observation
- Formulas of compounds must never be changed while balancing a chemical equation
 - Subscripts in a formula cannot be changed, nor can atoms be added or subtracted from a formula

Problem-Solving Strategy - Writing and Balancing the Equation for a Chemical Reaction

1. Determine what reaction is occurring
 - Determine the reactants, the products, and the physical states involved
2. Write the unbalanced equation that summarizes the reaction

Problem-Solving Strategy - Writing and Balancing the Equation for a Chemical Reaction

3. Balance the equation by inspection, starting with the most complicated molecule(s)
 - Determine what coefficients are necessary
 - Same number of each type of atom needs to appear on both reactant and product sides
 - Do not change the formulas of any of the reactants or products

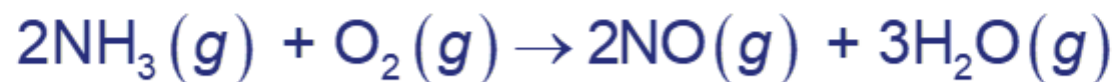
Example 1

- At 1000°C, ammonia gas, $\text{NH}_3(g)$, reacts with oxygen gas to form gaseous nitric oxide, $\text{NO}(g)$, and water vapor
 - This reaction is the first step in the commercial production of nitric acid by the Ostwald process
 - Balance the equation for this reaction

- Let's begin by balancing the hydrogen
 - A coefficient of 2 for NH_3 and a coefficient of 3 for H_2O give six atoms of hydrogen on both sides

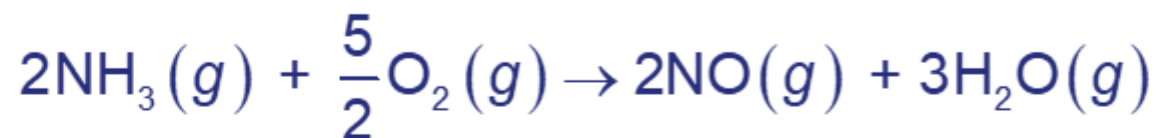


- Nitrogen can be balanced with a coefficient of 2 for NO



- Finally, note that there are two atoms of oxygen on the left and five on the right

- Oxygen can be balanced with a coefficient of $\frac{5}{2}$ for O_2



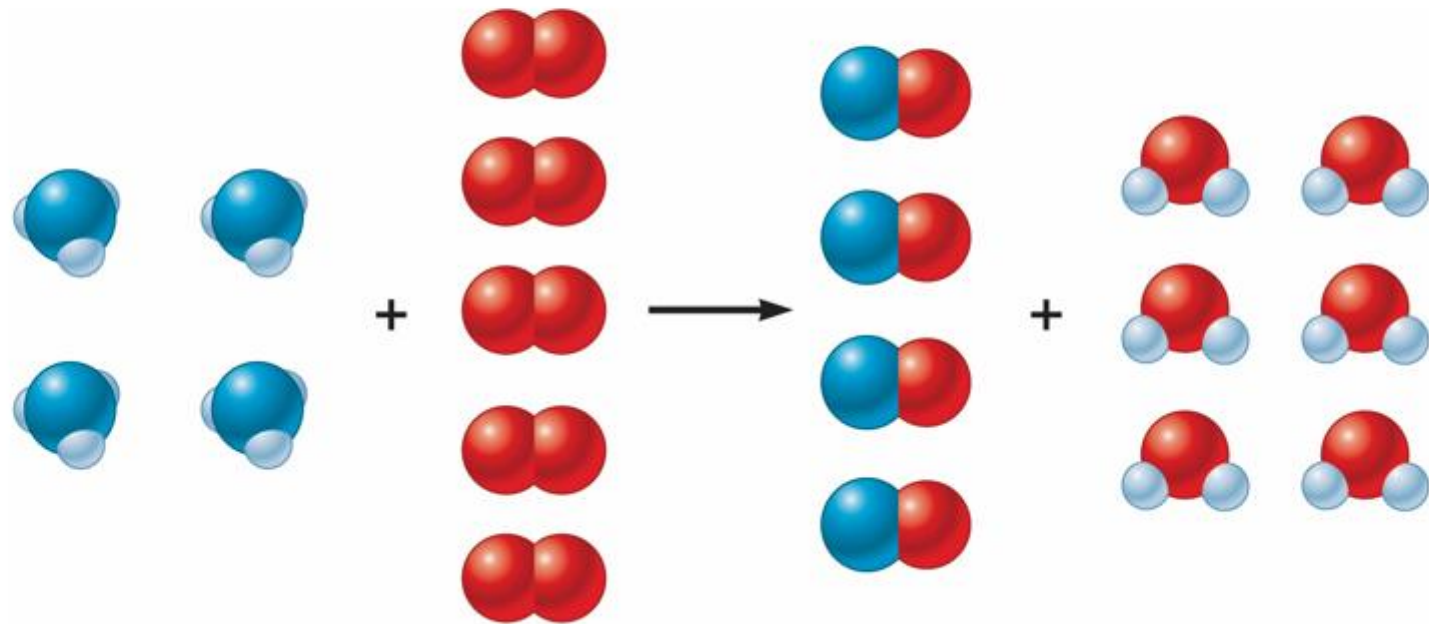
– Usual custom is to have whole-number coefficients

- We simply multiply the entire equation by 2

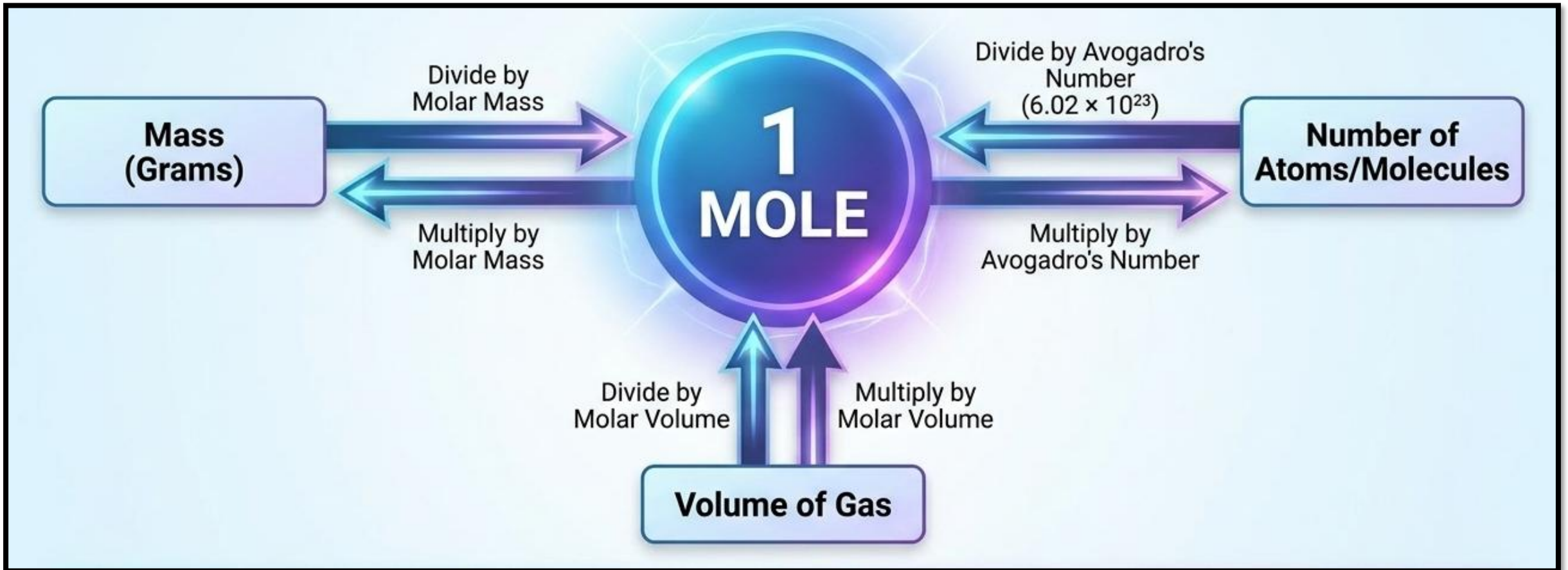


- Reality check - There are 4 N, 12 H, and 10 O on both sides, so the equation is balanced

- Visual representation of the balanced reaction

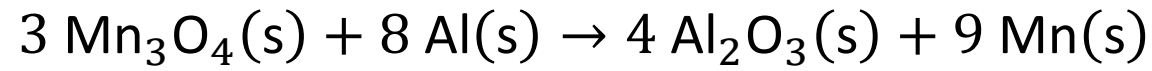


Stoichiometric calculation



Question 1

- At a very high temperature, manganese is isolated from its ore, manganomanganic oxide via the following balanced equation:



- a. How many manganese atoms are liberated if 62.4 moles of Mn_3O_4 react with excess aluminum?
- b. How many moles of aluminum oxide are made if 4250 g of manganomanganic oxide are consumed?
- c. How many moles of manganomanganic oxide will react with 2.85×10^{24} atoms of aluminum?
- d. If 5.12 moles of aluminum are consumed, how many molecules of aluminum oxide are produced?

Molar Mass Reference:

- $\text{Mn}_3\text{O}_4 = (3 \times 54.94 \text{ g/mol}) + (4 \times 16.00 \text{ g/mol}) = 228.82 \text{ g/mol}$
- Avogadro's Number (N_A) = 6.022×10^{23} particles/mol

a) Find the mole ratio: From the balanced equation, 3 moles of Mn_3O_4 produce 9 moles of Mn (a 3:1 ratio).

$$\text{Moles of Mn} = 62.4 \text{ mol Mn}_3\text{O}_4 \times \frac{9 \text{ mol Mn}}{3 \text{ mol Mn}_3\text{O}_4} = 187.2 \text{ mol Mn}$$

Convert moles to atoms: Multiply by Avogadro's number.

$$\text{Atoms of Mn} = 187.2 \text{ mol} \times (6.022 \times 10^{23} \text{ atoms/mol}) = 1.13 \times 10^{26} \text{ atoms}$$

b) Convert grams of Mn_3O_4 to moles:

$$\text{Moles of Mn}_3\text{O}_4 = \frac{4250 \text{ g}}{228.82 \text{ g/mol}} = 18.574 \text{ mol}$$

Apply the mole ratio: From the equation, 3 moles of Mn_3O_4 yield 4 moles of Al_2O_3 .

$$\text{Moles of Al}_2\text{O}_3 = 18.574 \text{ mol Mn}_3\text{O}_4 \times \frac{4 \text{ mol Al}_2\text{O}_3}{3 \text{ mol Mn}_3\text{O}_4} = 24.77 \text{ mol}$$

c) Convert atoms of Al to moles:

$$\text{Moles of Al} = \frac{2.85 \times 10^{24} \text{ atoms}}{6.022 \times 10^{23} \text{ atoms/mol}} = 4.733 \text{ mol}$$

Apply the mole ratio: From the equation, 8 moles of Al react with 3 moles of Mn_3O_4 .

$$\text{Moles of Mn}_3\text{O}_4 = 4.733 \text{ mol Al} \times \frac{3 \text{ mol Mn}_3\text{O}_4}{8 \text{ mol Al}} = 1.775 \text{ mol}$$

Answer: 1.77 moles

d) Apply the mole ratio: From the equation, 8 moles of Al produce 4 moles of Al_2O_3 (a 2:1 ratio).

$$\text{Moles of Al}_2\text{O}_3 = 5.12 \text{ mol Al} \times \frac{4 \text{ mol Al}_2\text{O}_3}{8 \text{ mol Al}} = 2.56 \text{ mol Al}_2\text{O}_3$$

Convert moles to molecules: Multiply by Avogadro's number.

$$\text{Molecules of Al}_2\text{O}_3 = 2.56 \text{ mol} \times (6.022 \times 10^{23} \text{ molecules/mol}) = 1.54 \times 10^{24} \text{ molecules}$$

Formula of a compound

Determining the Formula of a Compound - Process

- Weigh the sample of the compound by:
 - Decomposing the sample into its constituent elements
 - Reacting it with oxygen
- Resulting substances are then collected and weighed
 - This type of analysis is done by a combustion device
 - Results of such analyses provide the mass of each type of element in the compound, which can be used to determine the mass percent of each element

Empirical Formula

- Any molecule that can be represented as $(\text{CH}_5\text{N})_n$ has the empirical formula CH_5N
 - n - Integer
 - **Molecular formula:** Exact formula of the molecules present in a substance
 - Molecular formula = (empirical formula) $_n$
 - Requires the knowledge of the molar mass

Empirical Formula

The **simplest whole-number ratio** of elements.



Example: CH_5N

Molecular Formula

The **exact** blueprint of molecules present.
Requires knowing the molar mass.



Example: $(\text{CH}_5\text{N})_n$

1. Percent to Mass



Treat the sample as
100g total.

2. Mass to Moles



Divide by atomic mass
(Use the Mole Bridge).

3. Divide by Small



Divide all mole
values by the smallest
number to get a
relative ratio.

4. Multiply 'til Whole

$\times 3$

If you get a fraction
(like 3.33), multiply by
an integer to reach
whole numbers.

Example 2

Element	Mass %	Atomic Mass	Moles (Relative No.)	Simplest Ratio	Whole No. Ratio
C	40.0%	12.0	$\frac{40.0}{12.0} = 3.33$	$\frac{3.33}{3.33} = 1$	1
H	6.67%	1.0	$\frac{6.67}{1.0} = 6.67$	$\frac{6.67}{3.33} = 2$	2
O	53.33%	16.0	$\frac{53.33}{16.0} = 3.33$	$\frac{3.33}{3.33} = 1$	1

Question 2

- A compound contains 47.08% carbon, 6.59% hydrogen, and 46.33% chlorine by mass
 - Molar mass of the compound is 153 g/mol
 - What are the empirical and molecular formulas of the compound?

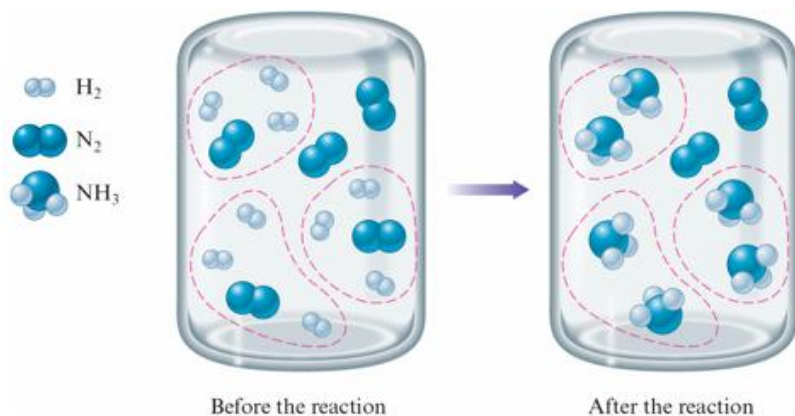
Empirical formula - $\text{C}_3\text{H}_5\text{Cl}$

Molecular formula - $\text{C}_6\text{H}_{10}\text{Cl}_2$

The concept of Limiting Reactant

Stoichiometric Mixture

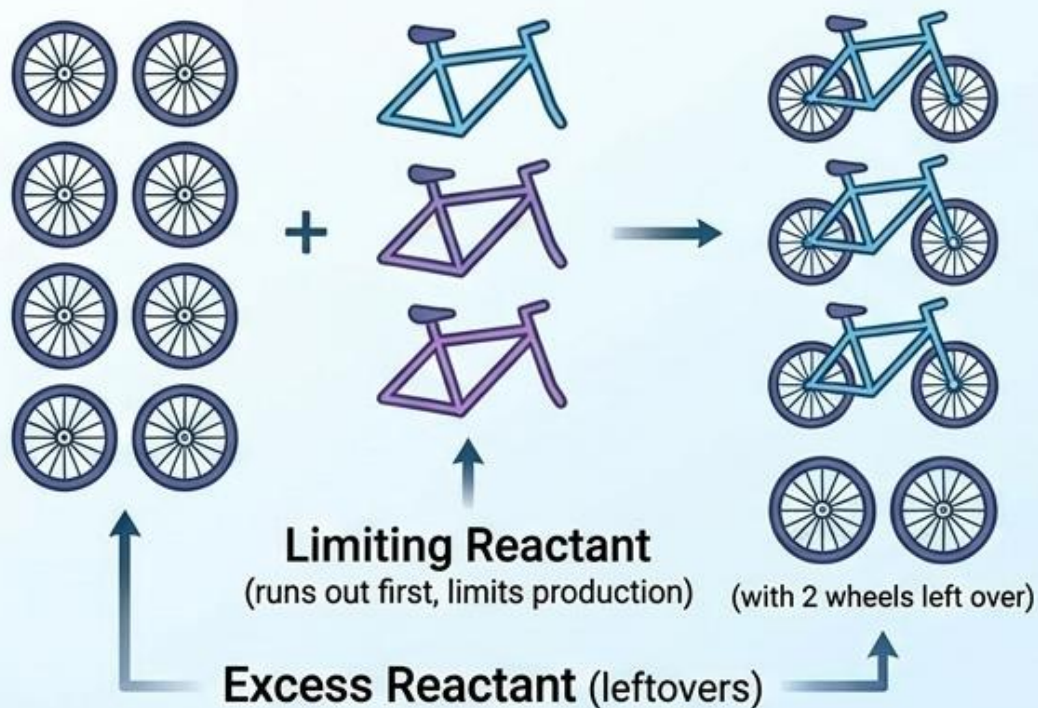
- A type of mixture that contains relative amounts of reactants that match the numbers in the balanced equation



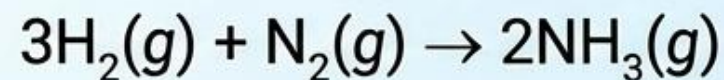
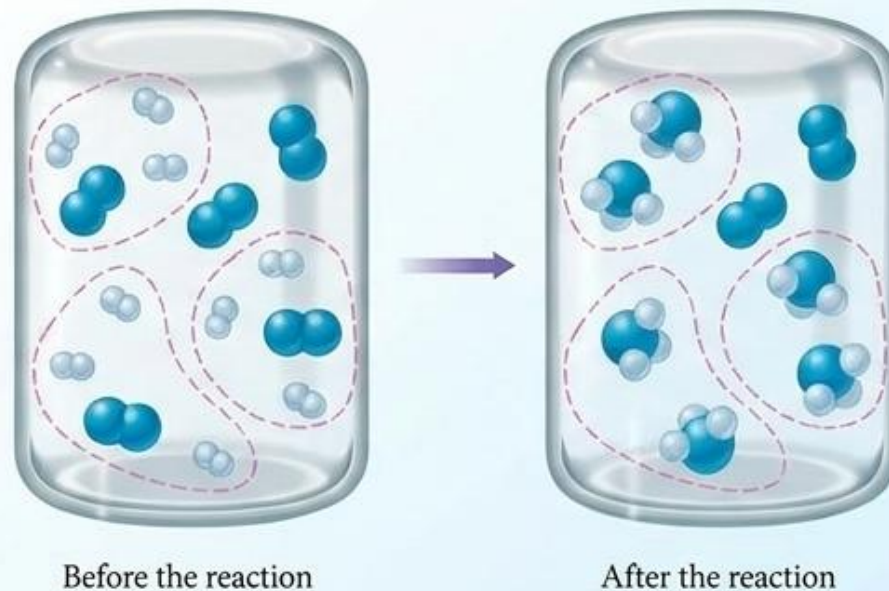
Limiting Reactant

- One that runs out first and thus limits the amount of product that can form
 - To determine how much product can be formed from a given mixture of reactants, one must look for the reactant that is limiting
- Some mixtures can be stoichiometric
 - All reactants run out at the same time
 - Requires determining which reactant is limiting

The Analogy



The Chemistry



To determine how much product can be formed from a mixture, you must ALWAYS find the limiting reactant first. The one that produces the smaller amount of product is the limit.

Example 3

- If a sample containing 18.1 g of NH_3 is reacted with 90.4 g of CuO , which is the limiting reactant?
 - How many grams of N_2 will be formed?
- What information do we need?
 - Balanced equation for the reaction
 - Moles of NH_3
 - Moles of CuO

Solution

Step 1

- To find the limiting reactant, determine the balanced equation



Step 2

- What are the moles of NH_3 and CuO ?
 - To find the moles, we need to know the molar masses

NH_3 17.03 g/mol

CuO 79.55 g/mol

$$18.1 \cancel{\text{ g NH}_3} \times \frac{1 \text{ mol NH}_3}{17.03 \cancel{\text{ g NH}_3}} = 1.06 \text{ mol NH}_3$$

$$90.4 \cancel{\text{ g CuO}} \times \frac{1 \text{ mol CuO}}{79.55 \cancel{\text{ g CuO}}} = 1.14 \text{ mol CuO}$$

Step 3

- What is the mole ratio between NH_3 and CuO in the balanced equation?

$$\frac{3 \text{ mol CuO}}{2 \text{ mol NH}_3}$$

Step 4

- How many moles of CuO are required to react with 1.06 moles of NH_3 ?

$$1.06 \text{ mol NH}_3 \times \frac{3 \text{ mol CuO}}{2 \text{ mol NH}_3} = 1.59 \text{ mol CuO}$$

- Thus 1.59 moles of CuO are required to react with 1.06 moles of NH_3
- Since only 1.14 moles of CuO are actually present, the amount of CuO is limiting; CuO will run out before NH_3 does

Or

Alternatively we can determine the limiting reactant by computing the moles of N_2 that would be formed by complete consumption of NH_3 and CuO

$$1.06 \cancel{\text{ mol NH}_3} \times \frac{1 \text{ mol N}_2}{2 \cancel{\text{ mol NH}_3}} = 0.530 \text{ mol N}_2$$

$$1.14 \cancel{\text{ mol CuO}} \times \frac{1 \text{ mol N}_2}{3 \cancel{\text{ mol CuO}}} = 0.380 \text{ mol N}_2$$

- As before, we see that the CuO is limiting since it produces the smaller amount of N_2
- To find the mass of N_2 produced, determine the moles of N_2 formed
 - Because CuO is the limiting reactant, we must use the amount of CuO to calculate the amount of N_2 formed
- What mass of N_2 is produced?
 - Using the molar mass of N_2 (28.02 g/mol), we can calculate the mass of N_2 produced

$$0.380 \cancel{\text{ mol N}_2} \times \frac{28.02 \text{ g N}_2}{1 \cancel{\text{ mol N}_2}} = 10.6 \text{ g N}_2$$

Concept of yield

- **Theoretical yield:** Amount of product formed when the limiting reactant is entirely consumed
 - Amount of product predicted is rarely obtained because of side reactions and other complications
- **Percent yield:** Actual yield of product
 - Often provided as a percentage of the theoretical yield

$$\frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\% = \text{percent yield}$$



Theoretical Yield

The amount of product formed when the limiting reactant is entirely consumed. A perfect, on-paper calculation.



Actual Yield

The real amount obtained in the lab. Rarely perfect due to side reactions, lost material, and complications.

Formula Bar: $\left(\frac{\text{Actual yield}}{\text{Theoretical yield}} \right) \times 100\% = \text{percent yield}$

Example 4

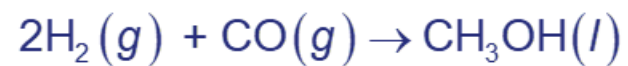
- Methanol can be manufactured by combining gaseous carbon monoxide and hydrogen

Suppose 68.5 kg CO(g) is reacted with 8.60 kg H₂(g)

- Calculate the theoretical yield of methanol
 - If 3.57×10^4 g CH₃OH is actually produced, what is the percent yield of methanol?
- What information do we need?
 - Balanced equation for the reaction
 - Moles of H₂
 - Moles of CO
 - Which reactant is limiting
 - Amount of CH₃OH produced

Step 1

- To find the limiting reactant, balance the chemical equation



Step 2

- To find the moles, we need to know the molar masses

H₂ 2.016 g/mol

CO 28.02 g/mol

$$68.5 \cancel{\text{ kg CO}} \times \frac{1000 \cancel{\text{ g CO}}}{1 \cancel{\text{ kg CO}}} \times \frac{1 \text{ mol CO}}{28.02 \cancel{\text{ g CO}}} = 2.44 \times 10^3 \text{ mol CO}$$

$$8.60 \cancel{\text{ kg H}_2} \times \frac{1000 \cancel{\text{ g H}_2}}{1 \cancel{\text{ kg H}_2}} \times \frac{1 \text{ mol H}_2}{2.016 \cancel{\text{ g H}_2}} = 4.27 \times 10^3 \text{ mol H}_2$$

Step 3

- We can determine the limiting reactant by calculating the amounts of CH_3OH formed by complete consumption of $\text{CO}(g)$ and $\text{H}_2(g)$:

$$2.44 \times 10^3 \cancel{\text{ mol CO}} \times \frac{1 \text{ mol CH}_3\text{OH}}{1 \cancel{\text{ mol CO}}} = 2.44 \times 10^3 \text{ mol CH}_3\text{OH}$$

$$4.27 \times 10^3 \cancel{\text{ mol H}_2} \times \frac{1 \text{ mol CH}_3\text{OH}}{2 \cancel{\text{ mol H}_2}} = 2.14 \times 10^3 \text{ mol CH}_3\text{OH}$$

- Since complete consumption of the H_2 produces the smaller amount of CH_3OH , the H_2 is the limiting reactant as we determined above

Step 4

- Determine the theoretical yield of CH_3OH in grams

$$2.14 \times 10^3 \cancel{\text{ mol CH}_3\text{OH}} \times \frac{32.04 \text{ g CH}_3\text{OH}}{1 \cancel{\text{ mol CH}_3\text{OH}}} = 6.86 \times 10^4 \text{ g CH}_3\text{OH}$$

Step 5

- Thus, from the amount of reactants given, the maximum amount of CH_3OH that can be formed is $6.86 \times 10^4 \text{ g}$
 - This is the theoretical yield
- Percent yield of CH_3OH

$$\frac{\text{Actual yield (grams)}}{\text{Theoretical yield (grams)}} \times 100 = \frac{3.57 \times 10^4 \text{ g CH}_3\text{OH}}{6.86 \times 10^4 \text{ g CH}_3\text{OH}} \times 100\% = 52.0\%$$

Solutions



The diagram shows a beaker filled with a blue liquid. Inside the liquid, there are many small blue spheres (solvent) and a few larger red spheres (solute). Three labels with lines pointing to the beaker are present: 'Solute' points to a red sphere, 'Solvent' points to a blue sphere, and 'Solution' points to the entire mixture. A bracket on the left side of the beaker also points to the entire mixture and is labeled 'Solution'.

Solute: The substance being dissolved.

Solvent: The dissolving medium (water, in aqueous solutions).

Solution: The homogenous mixture of both.

Solution: The homogenous mixture of both.

Percent by Mass

$$\left(\frac{\text{Mass of Solute}}{\text{Mass of Solution}} \right) \times 100$$

Percent by Volume

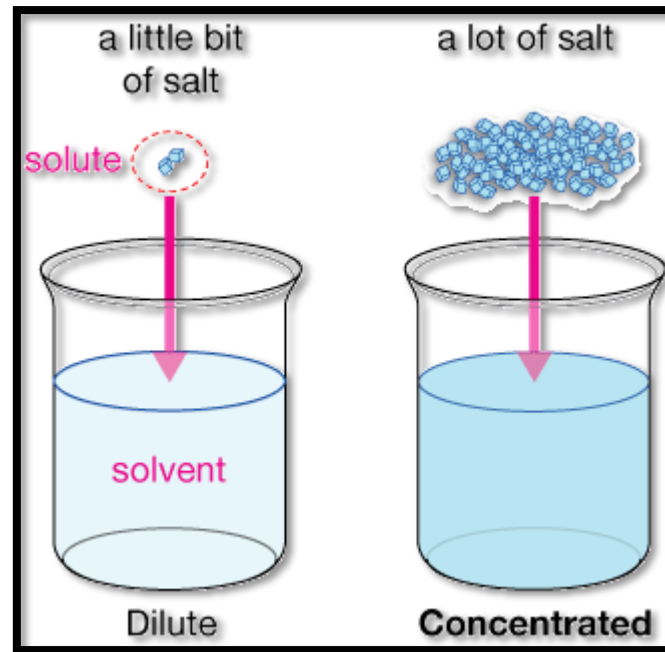
$$\left(\frac{\text{Volume of Solute}}{\text{Volume of Solution}} \right) \times 100$$

Standard Equation

$$n = \frac{MV}{1000}$$

(M = Molarity, V = Volume in mL)

Solutions



Concentration of solution

The concentration of a solution is the amount of solute present in a given quantity of solvent or solution.

Molarity	Molality
mol of solute / volume of solution	mol of solute/ mass of solvent
Units: $\text{mol L}^{-1}/\text{mol dm}^{-3}$	Unit: mol/kg
Symbol : M	Symbol:m
Affected by changes in Pressure & Temperature	Not affected by changes in Pressure & Temperature

Other Concentration unit

$$\text{Percent by Mass} = \frac{\text{Mass of Solute}}{\text{Mass of Solution}} \times 100$$

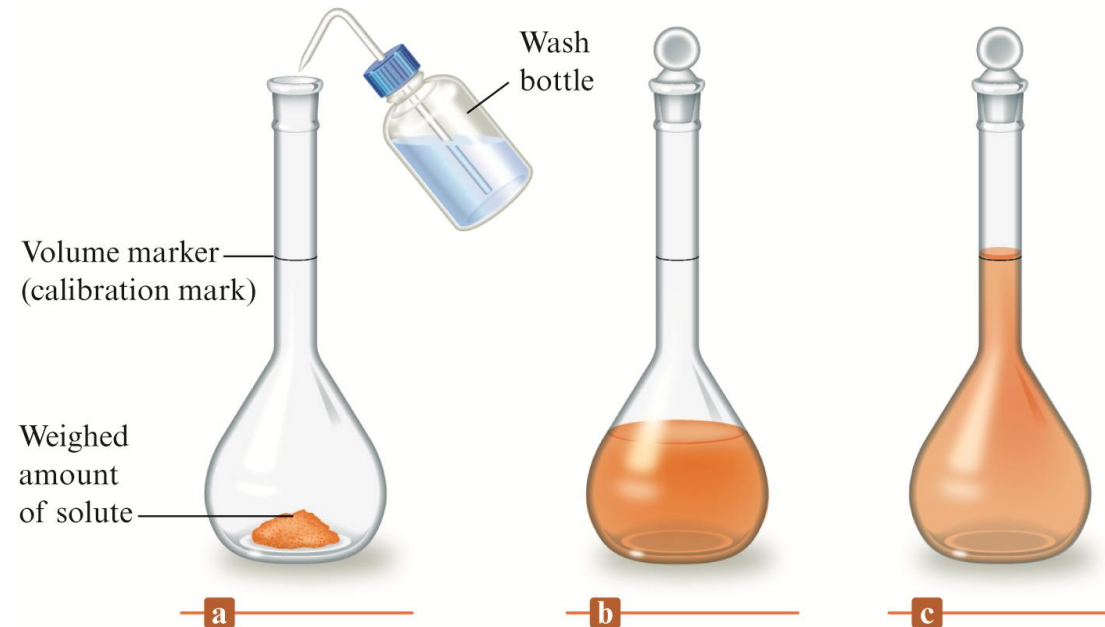
$$\text{Percent by Volume} = \frac{\text{Volume of Solute}}{\text{Volume of Solution}} \times 100$$

$$\text{mass/volume \% (m/v)} = \frac{\text{grams (g) of solute}}{\text{volume(mL) of solution}} \times 100\%$$

Preparation of standard solution

- Solution of a known concentration
- Preparation method
 - a. Weighing the solute

Use the equation
 $n = MV/1000$



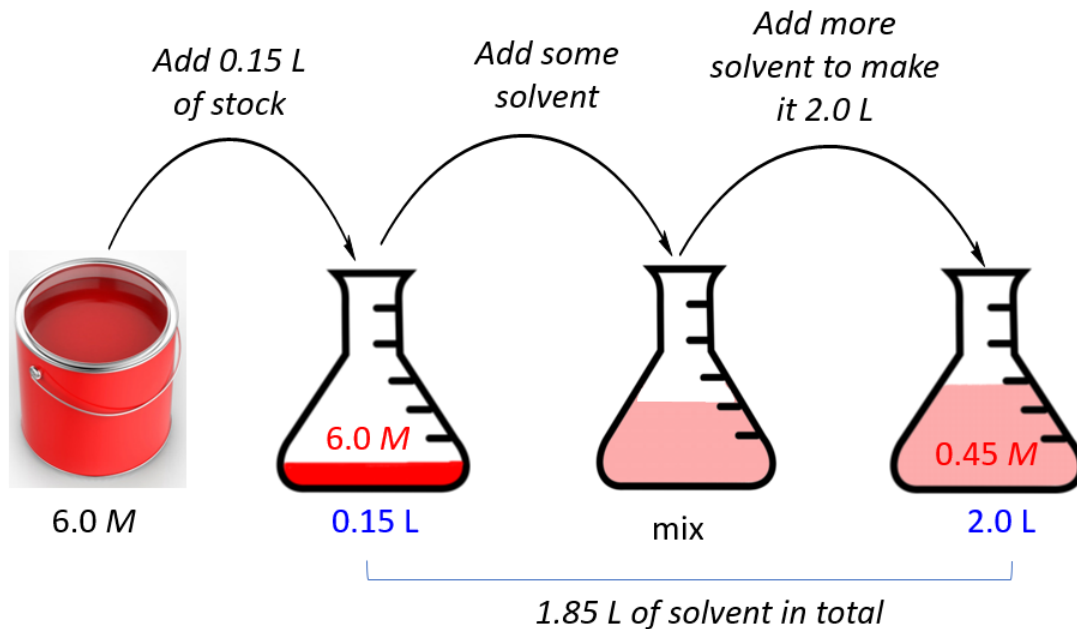
Preparation of standard solution

b. Dilution

procedure for preparing a less concentrated solution from a more concentrated solution

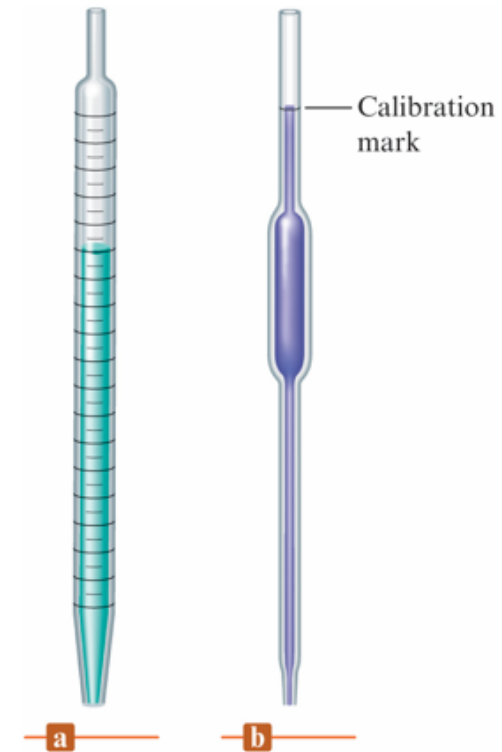
Use the equation

$$M_1V_1=M_2V_2$$



Glassware Used in Dilution

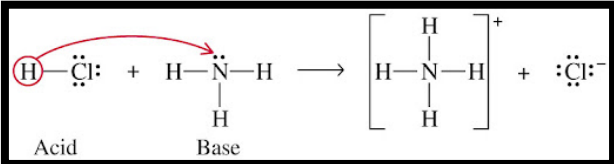
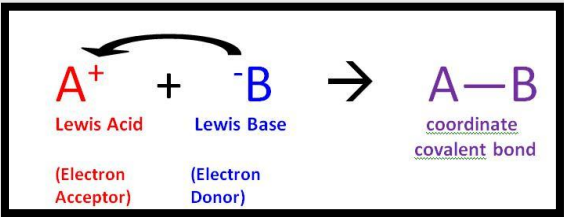
- Volumetric flask
- Pipet
 - Device used for the accurate measurement and transfer of a given volume of solution
 - Types
 - Volumetric or transfer pipet
 - Measuring pipet



Chemical reaction

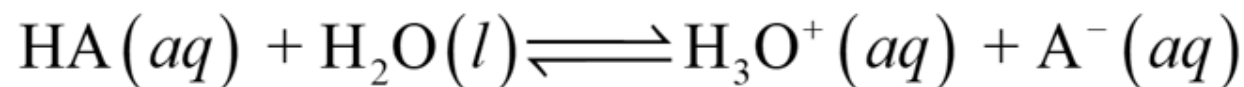
- Acid-base reaction
- Redox reaction

Theory of acid and base

Theory	Acid	Base	
Arrhenius	Acids produce hydrogen ions in aqueous solution	Bases produce hydroxide ions in aqueous solution	Acid: $\text{HCl} \xrightarrow{\text{H}_2\text{O}} \text{H}_3\text{O}^+ + \text{Cl}^-$ Base: $\text{NaOH} \xrightarrow{\text{H}_2\text{O}} \text{Na}^+ + \text{OH}^-$
Brønsted Lowry	Acids are proton donors	bases are proton (H^+) acceptors	
Lewis	Electron-pair acceptor <i>Contains an empty atomic orbital that can be used to accept an electron pair from the Lewis base</i> cations such as H^+ , Fe^{2+} , Al^{3+} molecules with incomplete octet central atom such as BF_3 , BeCl_2 Molecules with central atom that can expand octet such as PCl_3	Electron-pair donor <i>The species that can be Lewis base are: anions such as OH^-, CN^-, Cl^- molecules with lone pairs electrons at the central atom such as H_2O, NH_3, ROH molecules with nonpolar multiple bond such as alkenes</i>	

Acid in Water - General Reaction

- Reaction



Acid

Base

Conjugate
acid

Conjugate
base

- **Conjugate base:** Remains of an acid molecule after it loses a proton
- **Conjugate acid:** Formed when a proton is transferred to a base
- **Conjugate acid–base pair:** Two substances related to each other by accepting and donating a single proton

Nature of Acids

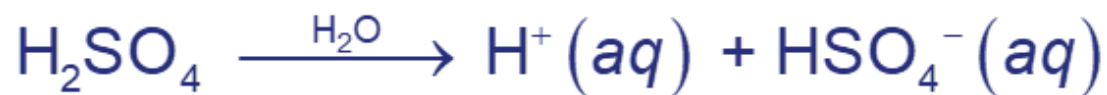
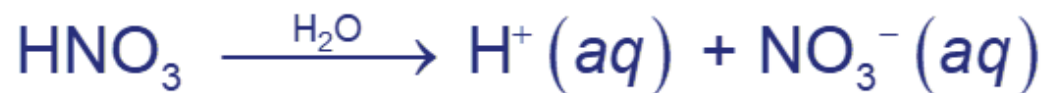
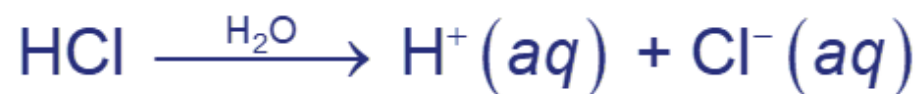
- Polar nature of water plays an important role in causing acids to produce H^+ ions
 - Ionization of an acid can be represented as follows:



- **Strong acids:** Every molecule is completely ionized when dissolved in water
 - Examples - HCl , HNO_3 , and H_2SO_4

Strong Acids

- Behave as strong electrolytes when dissolved in water



- **Acid:** Substance that produces H^+ ions when it is dissolved in water

Strong Bases

- Soluble ionic compounds that contain the hydroxide ion (OH^-)
 - When dissolved in water, cations and OH^- ions separate and move independently
- Example - Dissolving potassium hydroxide (KOH) in water



Weak Acids

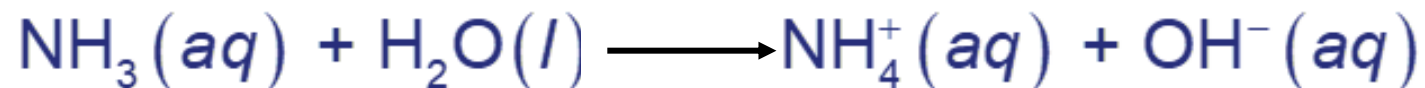
- Dissociate (ionize) only to a slight extent in aqueous solutions
- Example - Dissociation reaction for acetic acid in water



- Double arrow indicates that the reaction can occur in either direction

Weak Bases

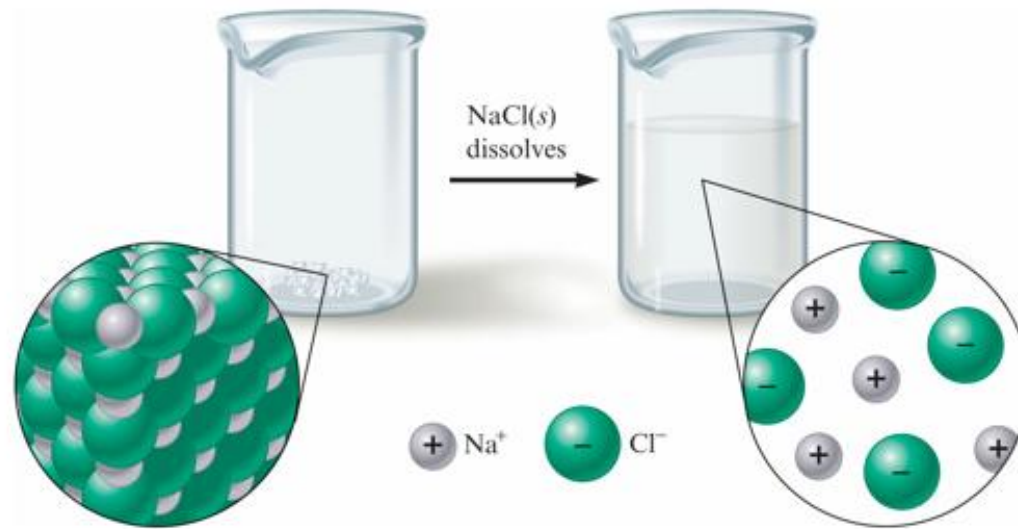
- Example - When NH_3 is dissolved in H_2O , it reacts as follows:



- Solution is basic because of the production of OH^- ions
- Double arrow indicates the reaction can occur in either direction

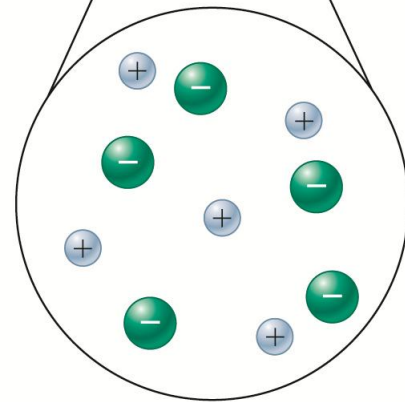
Soluble Salts

- Salts, such as NaCl, contain an array of cations and anions that disintegrate and undergo hydration when the salt dissolves

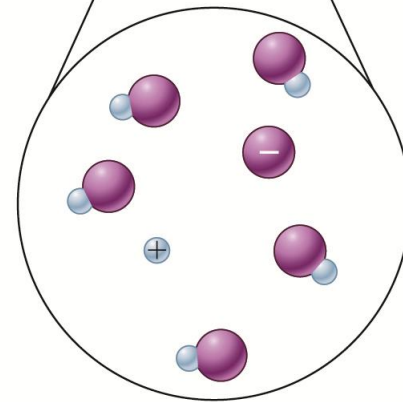


Electrical Conductivity of Aqueous Solution

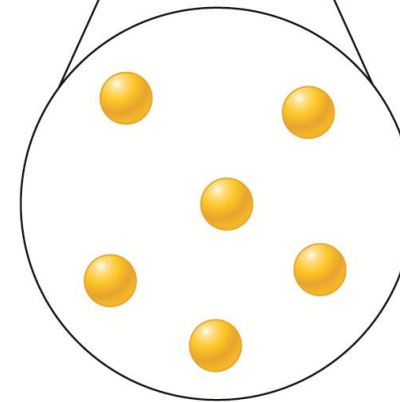
- Ability of a solution to conduct electric current
 - Pure water is not an electrical conductor
- Electrolyte
 - Substance that produces a solution that can conduct electricity when it is dissolved in water



a



b



c

Types of Electrolytes

- **Strong electrolytes:** Conduct current very efficiently
 - Example - NaCl
- **Weak electrolytes:** Conduct a small current in aqueous solutions
 - Example - Acetic acid
- **Nonelectrolytes:** Do not permit current to flow
 - Example - Sugar

Strong Electrolytes

- Substances that are completely ionized when they are dissolved in water
- Classes
 - Soluble salts
 - Strong acids
 - Strong bases

Weak Electrolytes

- Exhibit a small degree of ionization in water
- Include weak acids and weak bases
- Writing formulas for acids
 - Atom that produces H^+ ions (acidic hydrogen atom) in the solution is listed first
 - Nonacidic hydrogen atoms are written later (if present)

Nonelectrolytes

- Substances that dissolve in water but do not produce any ions
 - Resulting solution does not conduct an electric current
- Examples
 - Ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 - Table sugar (sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$)

Acid–Base Titrations

- **Volumetric analysis:** Technique used for ascertaining the amount of a certain substance by doing a titration
 - **Titration:** Delivery of a measured volume of a titrant into a solution containing an analyte
 - Titrant - Solution of known concentration
 - Analyte - Substance being analyzed

Acid–Base Titrations: Equivalence (Stoichiometric) Point

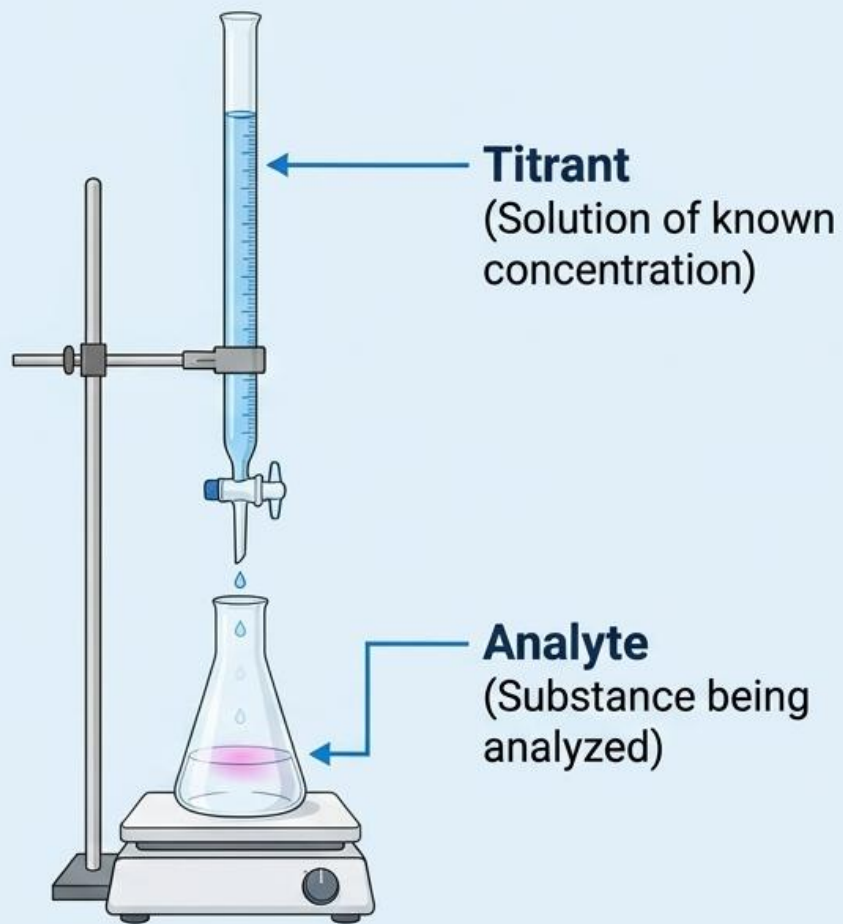
- Point in titration where enough titrant has been added to react exactly with the analyte
 - **Indicator:** Used to mark the equivalence point
 - Substance added at the beginning of the titration that changes color at the equivalence point
 - **Endpoint:** Point where the indicator actually changes color
 - Occurs exactly at the equivalence point

Requirements for a Successful Titration

- Exact reaction between titrant and analyte must be known and rapid
- Stoichiometric (equivalence) point must be marked accurately
- Volume of titrant required to reach the stoichiometric point must be known accurately

Acid–Base Titrations: Indicators

- Phenolphthalein
 - Common indicator
 - Colorless in an acidic solution and pink in a basic solution
 - Remains colorless when an acid is titrated with a base until after the acid is consumed and the first drop of excess base is added
 - Endpoint occurs approximately one drop of base beyond the stoichiometric point



Equivalence Point (The Invisible Math)

The exact stoichiometric point where enough titrant is added to perfectly react with the analyte.

Endpoint (The Visible Signal)

The point where the added Indicator actually changes color to mark the equivalence point.

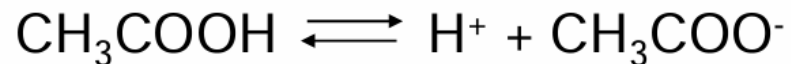
Monoprotic acids



Strong electrolyte, strong acid



Strong electrolyte, strong acid

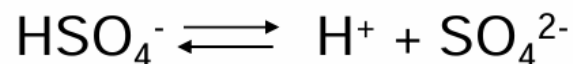


Weak electrolyte, weak acid

Diprotic acids



Strong electrolyte, strong acid



Weak electrolyte, weak acid

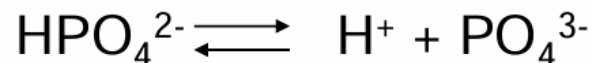
Triprotic acids



Weak electrolyte, weak acid



Weak electrolyte, weak acid



Weak electrolyte, weak acid

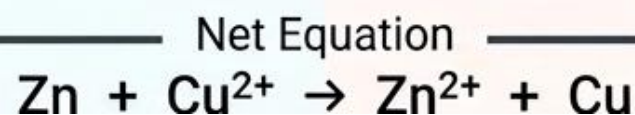
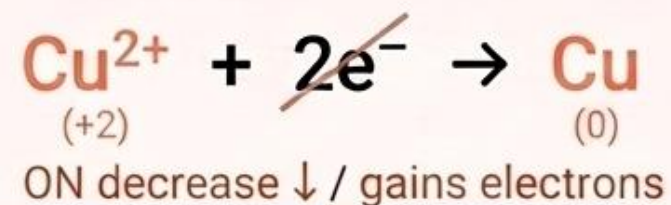
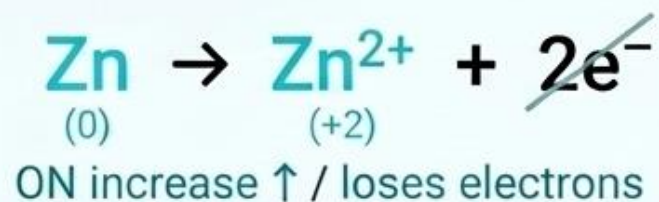
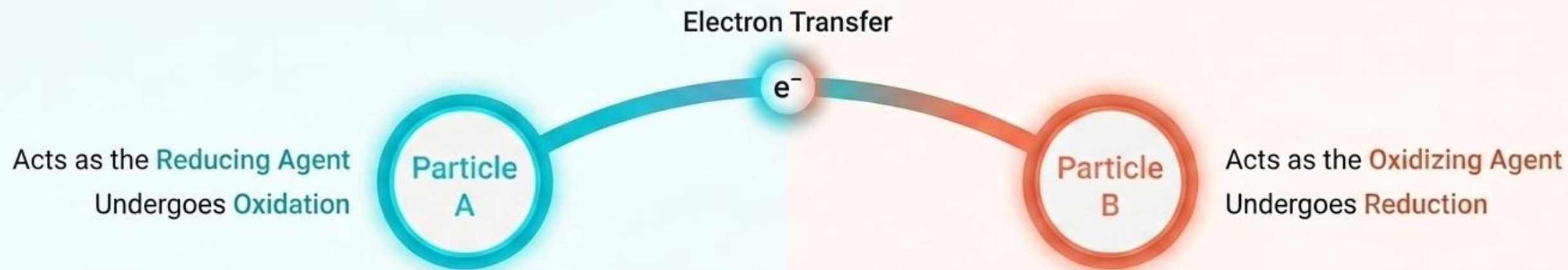
REDOX REACTION

Oxidation–Reduction (Redox) Reactions

- Involve the transfer of one or more electrons
- Example
 - Combustion reactions, which provide most of the energy to power human civilization



↓ Loss	Electrons (e ⁻)	↑ Gain
↓ Loss	Hydrogen (H)	↑ Gain
↑ Gain	Oxygen (O)	↓ Loss
↑ Increase	Oxidation Number (ON)	↓ Decrease



Oxidation States (Oxidation Numbers)

- For ionic compounds that contain monatomic ions, the oxidation states of the ions are equal to the ion charges
- For electrically neutral compounds, the sum of oxidation states must be zero
- For an ion, the sum of the oxidation states must equal the charge of the ion
- Written $+n$ or $-n$

The Oxidation State of . . .	Summary	Examples
An atom in an element is zero	Element: 0	Na(s), O ₂ (g), O ₃ (g), Hg(l)
A monatomic ion is the same as its charge	Monatomic ion: charge of ion	Na ⁺ , Cl ⁻
Fluorine is -1 in its compounds	Fluorine: -1	HF, PF ₃
Oxygen is usually -2 in its compounds Exception: peroxides (containing O₂²⁻), in which oxygen is -1	Oxygen: -2	H ₂ O, CO ₂
Hydrogen is +1 in its covalent compounds	Hydrogen: +1	H ₂ O, HCl, NH ₃

Question

- Assign oxidation states to all atoms in the following:
 - a. CO_2
 - b. NO_3^-

- Since each oxygen is -2 and there are two oxygen atoms, the carbon atom must be assigned an oxidation state of $+4$

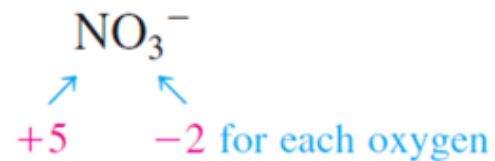


- Reality check

Number of C atoms = Number of O atoms

$$1(+4) + 2(-2) = 0$$

- Oxygen has an oxidation state of -2
 - Because the sum of the oxidation states of the three oxygens is -6 and the net charge on the NO_3^- ion is $1-$, the nitrogen must have an oxidation state of $+5$



- Reality check
 - $+5 + 3(-2) = -1$
 - Note that in this case the sum must be -1

Exercise

- Assign oxidation states for all atoms in each of the following compounds:



$$\text{K} = +1; \text{O} = -2; \text{Mn} = +7$$



$$\text{Na} = +1; \text{Fe} = +2; \text{O} = -2; \text{H} = +1$$



$$\text{H} = +1; \text{O} = -2; \text{N} = -3; \text{P} = +5$$



$$\text{C} = 0; \text{H} = +1; \text{O} = -2$$

Characteristics of Oxidation–Reduction Reactions

- Oxidation–reduction reactions are characterized by a transfer of electrons
- **Oxidation:** Increase in oxidation state
 - Characterized by loss of electrons
 - **Oxidizing agent:** Electron acceptor
- **Reduction:** Decrease in oxidation state
 - Characterized by gain of electrons
 - **Reducing agent:** Electron donor

Type of Oxidation and Reduction

Combustion reaction – It is a type of redox reaction which occurs between molecular oxygen and compound to form oxygen-containing products.



Disproportionation reaction – It is a type of redox reaction where a single reactant is reduced and oxidized. It is also known as an auto-oxidation reaction.



Single replacement reaction – It is a type of redox reaction that involves two elements switching places within a compound. It is also known as a single displacement reaction.

